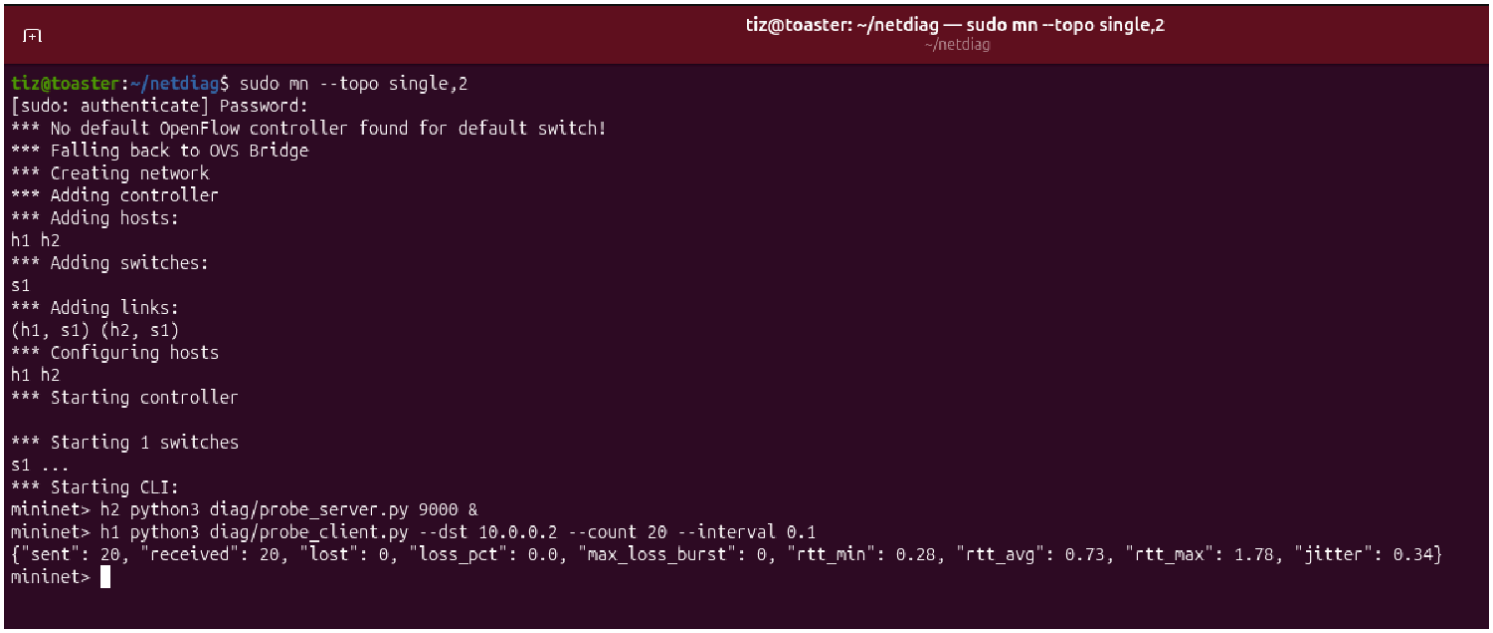
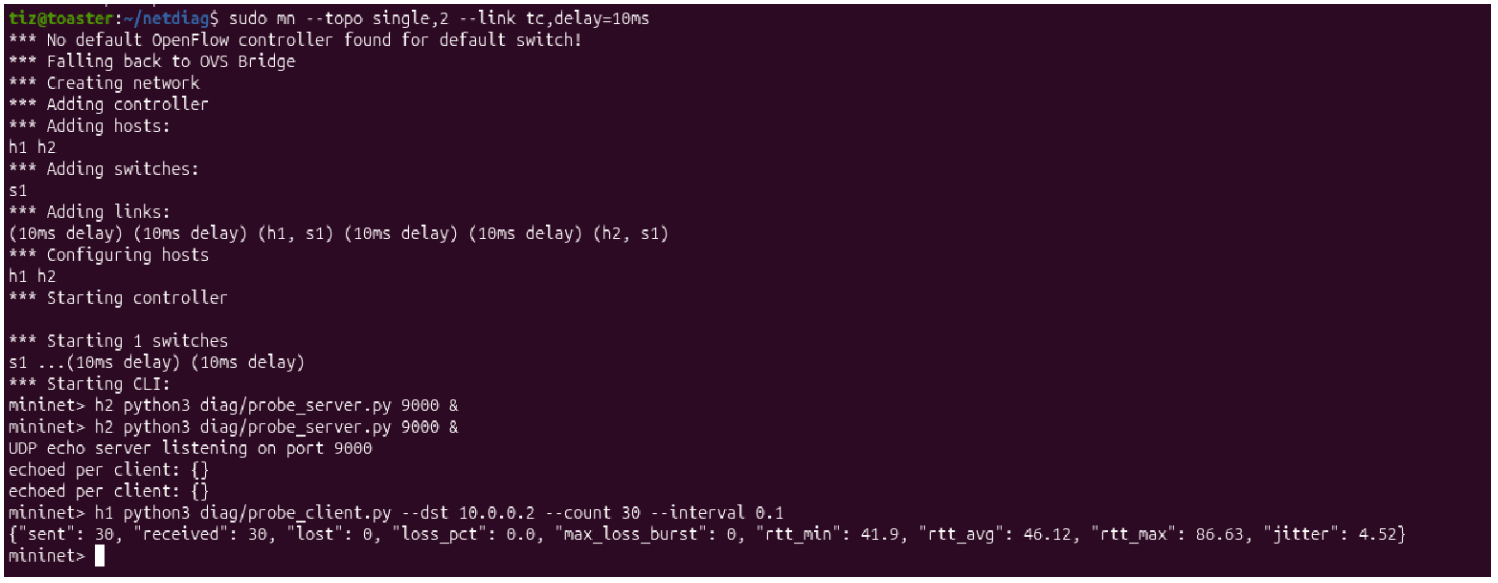
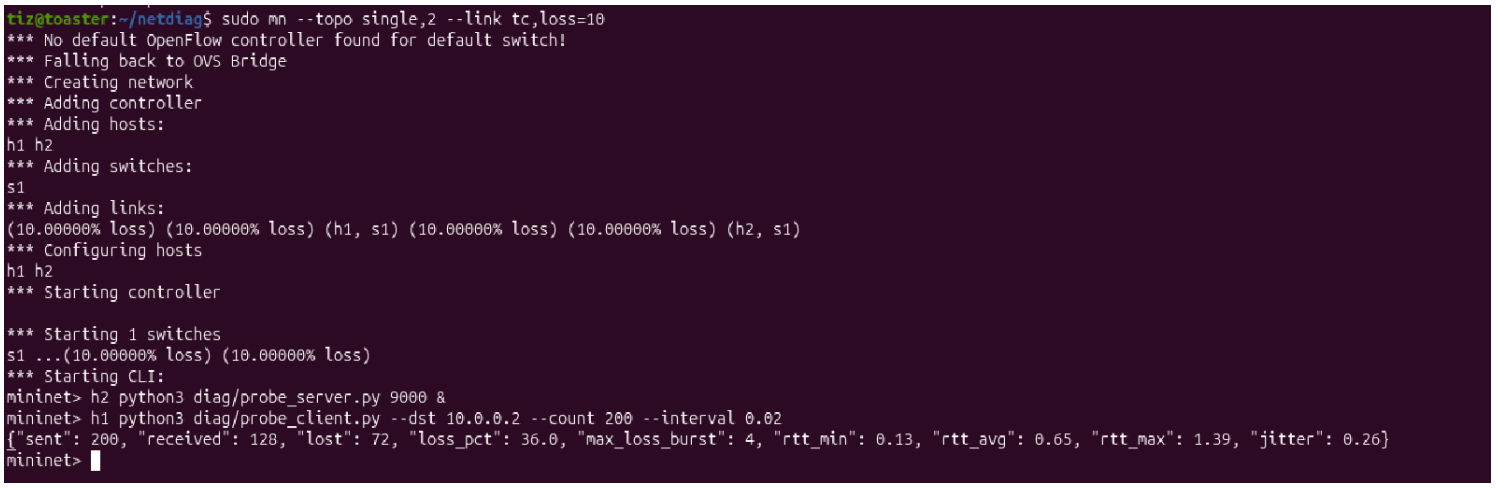
UDP


tiz@toaster: ~/netdiag — sudo mn --topo single,2
~/netdiag
tiz@toaster:~/netdiag$ sudo mn --topo single,2
[sudo: authenticate] Password:
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(h1, s1) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...
*** Starting CLI:
mininet> h2 python3 diag/probe_server.py 9000 &
mininet> h1 python3 diag/probe_client.py --dst 10.0.0.2 --count 20 --interval 0.1
{"sent": 20, "received": 20, "lost": 0, "loss_pct": 0.0, "max_loss_burst": 0, "rtt_min": 0.28, "rtt_avg": 0.73, "rtt_max": 1.78, "jitter": 0.34}
mininet>

RTT


tiz@toaster:~/netdiag$ sudo mn --topo single,2 --link tc,delay=10ms
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(10ms delay) (10ms delay) (h1, s1) (10ms delay) (10ms delay) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...(10ms delay) (10ms delay)
*** Starting CLI:
mininet> h2 python3 diag/probe_server.py 9000 &
mininet> h2 python3 diag/probe_server.py 9000 &
UDP echo server listening on port 9000
echoed per client: {}
echoed per client: {}
mininet> h1 python3 diag/probe_client.py --dst 10.0.0.2 --count 30 --interval 0.1
{"sent": 30, "received": 30, "lost": 0, "loss_pct": 0.0, "max_loss_burst": 0, "rtt_min": 41.9, "rtt_avg": 46.12, "rtt_max": 86.63, "jitter": 4.52}
mininet>

LOSS


tiz@toaster:~/netdiag$ sudo mn --topo single,2 --link tc,loss=10
*** No default OpenFlow controller found for default switch!
*** Falling back to OVS Bridge
*** Creating network
*** Adding controller
*** Adding hosts:
h1 h2
*** Adding switches:
s1
*** Adding links:
(10.00000% loss) (10.00000% loss) (h1, s1) (10.00000% loss) (10.00000% loss) (h2, s1)
*** Configuring hosts
h1 h2
*** Starting controller
*** Starting 1 switches
s1 ...(10.00000% loss) (10.00000% loss)
*** Starting CLI:
mininet> h2 python3 diag/probe_server.py 9000 &
mininet> h1 python3 diag/probe_client.py --dst 10.0.0.2 --count 200 --interval 0.02
{"sent": 200, "received": 128, "lost": 72, "loss_pct": 36.0, "max_loss_burst": 4, "rtt_min": 0.13, "rtt_avg": 0.65, "rtt_max": 1.39, "jitter": 0.26}
mininet>

TOPOLOGY


tiz@toaster:~/netdiag$ curl -s localhost:8080/topology | python3 -m json.tool
{
"switches": [
1,
2,
3,
4
],
"links": [
{
"src": "s1",
"dst": "s2",
"src_port": 3,
"dst_port": 1,
"util": 0.0
},
{
"src": "s1",
"dst": "s3",
"src_port": 4,
"dst_port": 1,
"util": 0.0
},
{
"src": "s2",
"dst": "s4",
"src_port": 2,
"dst_port": 3,
"util": 0.0
},
{
"src": "s3",
"dst": "s4",
"src_port": 2,
"dst_port": 4,
"util": 0.0
}
],
"hosts": [
{
"ip": "10.0.0.1",
"mac": "00:00:00:00:00:01",
"dpid": 1,
"port": 1
},
{
"ip": "10.0.0.2",
"mac": "00:00:00:00:00:02",
"dpid": 4,
"port": 1
},
{
"ip": "10.0.0.3",
"mac": "00:00:00:00:00:03",
"dpid": 1,
"port": 2
},
{
"ip": "10.0.0.4",
"mac": "00:00:00:00:00:04",
"dpid": 4,
"port": 2
}
]
}
tiz@toaster:~/netdiag$

PATH ANALISYS

```
tiz@toaster:~/netdiag$ curl -s "localhost:8080/path?src=10.0.0.1&dst=10.0.0.2" | python3 -m json.tool
{
    "src": "10.0.0.1",
    "dst": "10.0.0.2",
    "active_path": "s1-s2-s4",
    "pinned": null,
    "paths": [
        {
            "id": "s1-s2-s4",
            "switches": [
                1,
                2,
                4
            ],
            "hops": 3,
            "state": "UP",
            "metrics": null
        },
        {
            "id": "s1-s3-s4",
            "switches": [
                1,
                3,
                4
            ],
            "hops": 3,
            "state": "UP",
            "metrics": null
        }
    ]
}
```

FLOW TABLE

```
tiz@toaster:~/netdiag$ sudo ovs-ofctl -O OpenFlow13 dump-flows s1
[sudo: authenticate] Password:
 cookie=0x0, duration=370.082s, table=0, n_packets=811, n_bytes=48660, priority=65535,dl_dst=01:80:c2:00:00:0e,dl_type=0x88cc actions=CONTROLLER:65535
 cookie=0x0, duration=347.670s, table=0, n_packets=3, n_bytes=294, priority=100,ip,nw_src=10.0.0.1,nw_dst=10.0.0.2 actions=output:"s1-eth3"
 cookie=0x0, duration=347.670s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.2,nw_dst=10.0.0.1 actions=output:"s1-eth1"
 cookie=0x0, duration=347.629s, table=0, n_packets=3, n_bytes=294, priority=100,ip,nw_src=10.0.0.1,nw_dst=10.0.0.3 actions=output:"s1-eth2"
 cookie=0x0, duration=347.629s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.3,nw_dst=10.0.0.1 actions=output:"s1-eth1"
 cookie=0x0, duration=347.613s, table=0, n_packets=3, n_bytes=294, priority=100,ip,nw_src=10.0.0.1,nw_dst=10.0.0.4 actions=output:"s1-eth3"
 cookie=0x0, duration=347.612s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.4,nw_dst=10.0.0.1 actions=output:"s1-eth1"
 cookie=0x0, duration=347.547s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.2,nw_dst=10.0.0.3 actions=output:"s1-eth2"
 cookie=0x0, duration=347.547s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.3,nw_dst=10.0.0.2 actions=output:"s1-eth3"
 cookie=0x0, duration=347.422s, table=0, n_packets=3, n_bytes=294, priority=100,ip,nw_src=10.0.0.3,nw_dst=10.0.0.4 actions=output:"s1-eth3"
 cookie=0x0, duration=347.422s, table=0, n_packets=4, n_bytes=392, priority=100,ip,nw_src=10.0.0.4,nw_dst=10.0.0.3 actions=output:"s1-eth2"
 cookie=0x0, duration=370.090s, table=0, n_packets=91, n_bytes=8844, priority=0 actions=CONTROLLER:65535
```

CORRELATION

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py correlate
s1-s2-s4: avg RTT 27.01 ms (expected about 20), loss 0.0%
s1-s2-s4: avg RTT 26.72 ms (expected about 20), loss 0.0%
s1-s2-s4: avg RTT 23.8 ms (expected about 20), loss 0.0%
s1-s2-s4: avg RTT 25.21 ms (expected about 20), loss 0.0%
s1-s2-s4: avg RTT 24.87 ms (expected about 20), loss 0.0%
switch counters moved: {'s1': 27, 's2': 27, 's3': 0, 's4': 28}
```

FAILURE

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py failure
FAILURE RESULT: {'outage_s': 0.5, 'lost_probes': 3, 'detect_s': 0.02, 'reroute_s': 0.021}
```

FAILURE - - STATIC

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py failure --static
FAILURE RESULT: {'outage_s': 14.11, 'lost_probes': 28, 'detect_s': 0.018, 'reroute_s': None}
```

COMPARE

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py compare
condition path         avg RTT  max RTT  jitter  loss%
idle      s1-s2-s4       24.14    32.87    2.27    0.0
idle      s1-s3-s4       84.74     93.1    2.75   5.42
loaded    s1-s2-s4       27.29   219.01    5.33  30.83
loaded    s1-s3-s4       86.38   108.44    4.29    2.5
```

CONGESTION DYNAMIC

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py congestion
CONGESTION: {'sent': 500, 'loss_pct': 11.2, 'rtt_avg': 48.35, 'rtt_max': 450.86, 'jitter': 13.59} path changes: 2
```

CONGESTION STATIC

```
tiz@toaster:~/netdiag$ sudo python3 diag/analyzer.py congestion --static
CONGESTION: {'sent': 500, 'loss_pct': 10.2, 'rtt_avg': 25.59, 'rtt_max': 237.52, 'jitter': 3.52} path changes: 0
```